HarvestLink Rwanda: Cooperative Inventory Management Process

One-Page Business Process Explanation

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1. How the Business Process Begins and Progresses

The process begins when a cooperative member (farmer) initiates a delivery and hands over harvested produce. The Receiving Department inspects and weighs the produce, then records the transaction into the HarvestLink Oracle database. Once logged, the Inventory Department transfers the produce into physical storage. On a scheduled basis triggered by a periodic timer event, inventory staff conduct a physical stock count and compare those figures against the system-recorded stock. After discrepancies and post-harvest losses are evaluated and logged, a scheduled synchronization event updates the Power BI reporting layer, allowing Cooperative Management to review analytical reports and conclude the cycle.

2. The Actors Involved and Their Responsibilities

- **Cooperative Member (Farmer):** Initiates the workflow by delivering harvested produce to the cooperative.
- **Receiving Department:** Inspects produce quality, measures weight, and enters verified intake records into the Oracle database.
- **Inventory Department:** Transfers produce to storage, conducts periodic physical counts, reconciles physical versus recorded stock, investigates variances, and documents post-harvest losses in the system.
- **Cooperative Management (Power BI Viewer):** Reviews synchronized analytical reports and dashboards to oversee cooperative performance and conclude the reporting cycle.

3. How Decisions Are Made and Alternative Paths Are Handled

Decision logic is governed by an Exclusive Gateway ('Discrepancy Found?') following the comparison of physical and recorded stock:

- **Discrepancy Found (Yes Path):** If physical stock differs from recorded stock, the workflow branches to investigate the variance, identify post-harvest losses (such as spoilage or shrinkage), and record those losses in the Oracle database before feeding into the reporting sync.
- **No Discrepancy Found (No Path):** If physical and recorded quantities match, the investigation and loss-documentation activities are bypassed, and the process proceeds directly toward the scheduled Power BI synchronization.

4. How Information Moves Between Actors and Activities

Information flows sequentially through structured data artifacts across swimlanes. Initial delivery details and weights are captured as Produce Records during intake and written into the central Oracle database. During periodic audits, Inventory Data is retrieved from the database to verify physical counts. If variances occur, Loss Reports are generated and saved back into the Oracle database. Finally, a scheduled timer triggers an automated data sync message ('Dashboard Updated'), transferring consolidated inventory and loss metrics to Cooperative Management's Power BI interface.

5. How the Process Supports Organizational Operations and Management Decisions

Operationally, the process standardizes produce intake, enforces accountability across departments, and creates an auditable trail for stock movements and post-harvest losses. Strategically, automated Oracle-to-Power BI integration gives Cooperative Management timely visibility into stock discrepancies, spoilage trends, and net available inventory, enabling data-driven decisions on storage improvements, loss reduction, and sales planning.